

# Imitator: Direct Prediction of Gemma Token IDs for Language-Model-Assisted Sign Transcription

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**Abstract.** We present IMITATOR, a sign-transcription architecture designed to connect a lightweight visual model to a pretrained language model through its native token interface. The original IMITATOR proposal imitated continuous language-model embeddings. We instead predict explicit Gemma token IDs: this removes the ambiguity of decoding an approximate vector, permits exact sequence evaluation, and gives a downstream Gemma model a standard textual input that it can punctuate, normalize, or—when context permits—correct. A spatial-temporal keypoint encoder and autoregressive decoder are trained on synthetic streams of two to eight LSA64 signs and evaluated with signer-held-out splits. On three development signers, the selected model improves strict token-sequence accuracy from 1.8–2.3% for the initial autoregressive decoder to 16.9–19.8% while preserving the exact Gemma IDs. Prospectively registered evaluation on two additional signers obtains 12.5% and 14.1% strict accuracy. We also audit the complete token-to-language interface. Gemma 3n preserves 95% of clean isolated-label inputs, but recovers no exact sequence at any tested nonzero substitution rate and slightly decreases chrF. This negative result identifies a current boundary rather than a hidden language-model rescue: random isolated signs contain almost no sentence context from which to infer a corrupted word. The present evidence therefore supports signer-independent, LLM-compatible token transcription on controlled sign streams; correction of recognition errors in natural sentences remains a testable next-stage hypothesis.

**Keywords:** Sign language recognition · LLM tokens · ST-GCN · signer-independent evaluation

## 1 Introduction

Sign-language translation requires extracting linguistic evidence from motion and turning it into text. LLMs can address the latter, but visual encoders need a well-defined interface. The first published IMITATOR mapped 2D keypoints to a frozen LLM’s token-embedding space and reported preliminary MSE-cosine loss near  $8 \times 10^{-4}$  [18]. Yet geometric proximity does not guarantee the intended token, particularly in anisotropic contextual spaces [5], and its splits were not signer-exclusive.

We preserve the premise—a compact model learns the input language of a frozen LLM—but predict exact Gemma tokenizer IDs. Outputs are directly measurable and can enter the ordinary Gemma 3n text interface [7]. IMITATOR performs visually grounded transcription; Gemma performs linguistic realization only when sufficient context exists.

Our synthetic streams contain isolated signs in random order. This prevents a language-order shortcut but leaves little context for correcting a replaced word. We therefore score visual transcription and downstream correction separately. Our contributions are: (1) direct autoregressive prediction of Gemma IDs; (2) signer-disjoint LOSO evaluation with two prospectively held-out signers; (3) an explicit corruption audit showing no recovery at tested nonzero rates; and (4) causal validation of visual ordering through segment permutation.

## 2 Related Work

ST-GCN models skeleton motion naturally [21]. Recent skeleton CSLR combines joint groups, motion, temporal lift, and multi-scale features while highlighting signer diversity as a bottleneck [12,13,19]. Manual/non-manual cues, handshape boundaries, and feature attention provide complementary evidence [9,23,22], although small Transformers readily overfit [10].

Sign2GPT uses pseudo-gloss pretraining, SignLLM constructs hierarchical sign tokens, and FLA-LLM separates visual and language learning [20,6,2]; ST-GCN recognition has also been coupled to a language generator [14]. IMITATOR instead emits native tokenizer IDs inspectable without an LLM-side adapter.

Speech systems use continuous prompts or CTC token embeddings [3,11]. CTC is standard for monotonic alignment [8] and appears in CSLR and joint translation [1,16]. Our token-level CTC auxiliary did not improve development results, so the final interface is causal decoding.

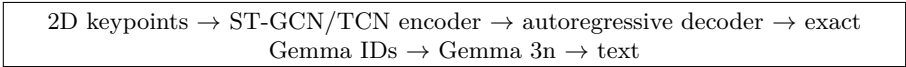
## 3 Discrete Token-ID Imitation

The present work is a direct successor to the published embedding-based IMITATOR [18], not a separate use of the same name. It preserves the modular premise—a small visual network learns the input language of a frozen LLM—while replacing the learned interface and the evaluation protocol.

Let  $X = (x_1, \dots, x_T)$  be a sequence of 2D keypoints and let a frozen tokenizer map the target text  $s$  to IDs  $Y = (y_1, \dots, y_L)$ . The original design learned vectors  $\hat{E} = f_\theta(X)$  and minimized a combination of mean-squared distance and cosine distance to frozen LLM embeddings  $E(Y)$ . Its downstream requirement was implicit: an approximate vector still had to be interpreted correctly by the language model.

The revised design makes that requirement the supervised objective:

$$\mathcal{L}_{\text{ID}} = - \sum_{i=1}^{L+1} \log p_\theta(y_i \mid y_{<i}, X), \quad y_{L+1} = \text{EOS}. \quad (1)$$



**Fig. 1.** The revised IMITATOR interface. Instead of approximating an LLM embedding, the visual model emits explicit tokenizer IDs. The language-model stage receives ordinary text tokens and is evaluated separately from visual transcription.

The prediction is correct only if the emitted ID sequence matches the tokenizer output and terminates with EOS. Thus, geometric alignment is replaced by a categorical, falsifiable interface contract. This does not make training annotation-free: in the present study, the text targets are derived from the 64 LSA64 sign labels. The precise claim is that no intermediate 64-class gloss classifier is required at inference, not that the supervision is gloss-free.

Figure 1 summarizes the intended two-stage system. The solid path is evaluated here. The final correction arrow is also audited, but current isolated-label data do not provide the sentence context needed to demonstrate error recovery.

## 4 Model

*Visual encoder.* Three ST-GCN blocks process each keypoint graph, followed by a 128-channel projection, node averaging, two temporal convolutions, and a one-layer, four-head Transformer. Fixed sinusoidal position is added before self-attention [17]. Without it, segment permutation left 86.3% of predictions unchanged; with it, content can be bound to video position. ST-GCN and the spatial projection use signer-clean initialization and remain frozen; temporal layers unfreeze after five epochs.

*Token decoder.* A two-layer, four-head causal Transformer (hidden size 128; feed-forward size 512) attends to visual memory. Input and output embeddings are tied. Greedy decoding starts at BOS, emits at most 32 content tokens, and stops at EOS. The task contains 118 Gemma content IDs; with BOS, EOS, and PAD, a bijection gives a 121-row softmax. Rows map back to the original IDs, so this is computational reparameterization rather than a new sign vocabulary.

*Language stage.* The same tokenizer decodes emitted IDs for a fixed Gemma 3n E2B instruction prompt. Gemma is not jointly fine-tuned. We report transcription before the LLM and correction afterward, preventing fluent output from masking visual error.

## 5 Data and Protocol

LSA64 has 3,200 videos of 64 Argentinian Sign Language signs, performed five times by ten subjects [15]. We normalize 111 upper-body, hand, and facial 2D keypoints per frame. Each example concatenates two to eight clips with optional

neutral gaps; its target concatenates the Gemma tokenizations of their Spanish labels. Random order prevents a linguistic next-sign prior but contains neither syntax nor coarticulation. The task is controlled token transcription, not continuous SLT.

Each LOSO fold excludes an outer-test signer, uses another for inner validation, and trains on the rest. Folds 1–6 are development, 7–8 prospective confirmation, and 9–10 reserved. E1 uses seed 23, 2,048 training examples per epoch, 512 fixed inner-validation examples, 30 epochs, and 896 outer examples. Checkpoint selection uses inner-validation edit similarity, then strict accuracy; outer results select nothing.

The preregistration requires, on each confirmation fold: more strict-accuracy wins than losses against historical affine CIF with sign-test  $p < .01$  (H1); strict accuracy  $\geq 11.9\%$  (H2); and pair order  $\geq .90$  plus permutation edit drop  $\geq .15$  (H3). CIF is only the frozen comparator and source initialization.

## 6 Evaluation

*Upstream transcription.* Strict accuracy is one only when all content IDs match and EOS is emitted. Token edit similarity is

$$\text{EditSim}(\hat{Y}, Y) = \max \left( 0, 1 - \frac{d_{\text{Lev}}(\hat{Y}, Y)}{|Y|} \right). \quad (2)$$

We report percentile bootstrap 95% confidence intervals with 10,000 fixed-seed replicates [4]. Paired E1-versus-CIF strict outcomes use an exact two-sided sign test after removing ties.

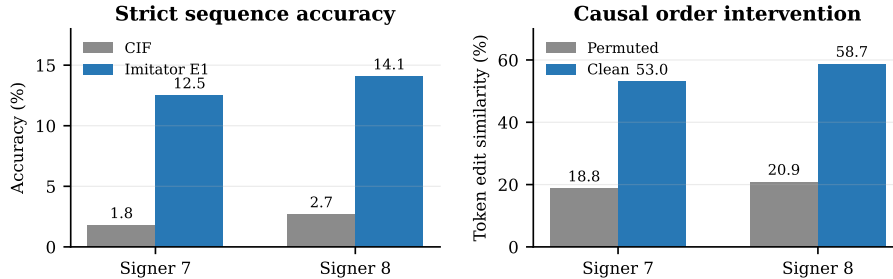
For the order diagnostic, token sequences are greedily parsed into known label tokenizations by longest match. A label is eligible if it occurs exactly once in prediction and target; all eligible pairs are scored for relative-order agreement. Samples with fewer than two eligible labels add no pair. We also permute complete input segments using gold synthetic boundaries, keep the target fixed, and measure the paired drop in edit similarity.

*Keypoint interpretation.* For one unseen-signer sequence, we compute the absolute Input $\times$ Gradient of each target-token log-probability with respect to every coordinate, normalize it per token, and aggregate keypoints into pose, face, left hand, and right hand. As a complementary intervention, each region is zeroed and the resulting decrease in target log-probability is measured. Gradients describe local sensitivity; ablation tests whether the region is functionally necessary for that example.

*Downstream correction audit.* To isolate the language stage from visual errors, we corrupt gold tokenizer sequences at fixed rates and submit the decoded inputs to Gemma with an explicit correction prompt. The corrected protocol substitutes complete one-token labels with other real labels; it does not insert arbitrary subword fragments. Labels that tokenize to one ID are excluded because replacing

**Table 1.** Prospectively registered signer-held-out confirmation. The CIF column is the frozen historical comparator, not a candidate IMITATOR architecture.

| Fold | CIF exact | E1 exact | E1 edit | Order | $\Delta_{\text{perm}}$ | W/L/T     | $p$      |
|------|-----------|----------|---------|-------|------------------------|-----------|----------|
| 7    | 1.8       | 12.5     | 53.0    | 98.2  | 0.343                  | 99/3/794  | 6.98e-26 |
| 8    | 2.7       | 14.1     | 58.7    | 98.9  | 0.379                  | 108/6/782 | 2.72e-25 |

**Fig. 2.** Confirmatory evidence on unseen signers. Left: strict sequence accuracy against the historical CIF comparator. Right: edit similarity collapses when complete sign segments are permuted while targets remain fixed.

their sole ID destroys all recoverable information. The audit contains 120 rows at each rate. Importantly, most remaining multi-token labels are single words split by the tokenizer; only five of the 64 labels are genuine multiword expressions. We measure exact decoded text and chrF before and after Gemma.

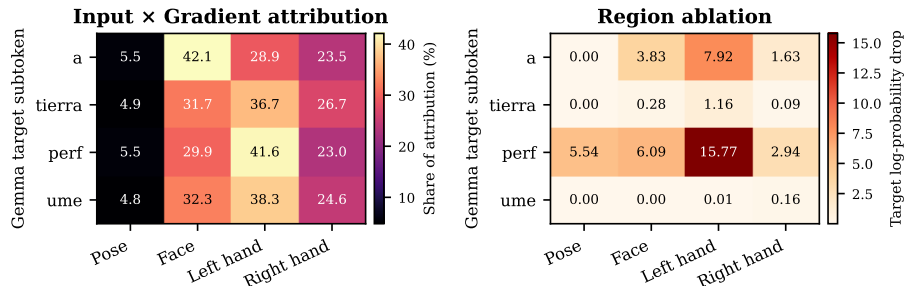
An oracle control on uncorrupted continuous Spanish reaches chrF 92.86 and content-word F1 0.985 after surface realization. It verifies preservation of well-formed content, not correction of recognition errors.

## 7 Results

### 7.1 Token-ID transcription

On development folds 4–6, E1 obtains 18.1, 19.8, and 16.9% strict accuracy, versus 2.2, 2.3, and 1.8% for the initial autoregressive model. Direct token-ID prediction is therefore viable but not solved: E1 improves strict sequence accuracy by roughly an order of magnitude over the initial autoregressive decoder, yet most complete sequences still contain at least one error. The positional intervention changes pairwise label order from near chance to approximately 0.99, supporting the claim that the emitted sequence is visually ordered.

Under the registered conjunctive decision rule, H1 passes, H2 passes, and H3 passes. Overall confirmation therefore passes. These statements are populated automatically from the frozen evaluation artifacts and are not manually reinterpreted after either fold.



**Fig. 3.** Qualitative keypoint interpretation for the target subtokens of an unseen signer-7 “A tierra / Perfume” sequence. Left: normalized Input $\times$ Gradient attribution aggregated into anatomical regions. Right: decrease in target log-probability when each region is zeroed. Both analyses identify the left hand as the dominant evidence for the central subtokens; this single example is descriptive rather than population-level evidence.

Figure 3 shows that the left hand contributes 28.9–41.6% of normalized attribution and causes the largest ablation loss for the central “a” and “perf” decisions (7.92 and 15.77 log-probability points). Face keypoints also receive substantial local attribution, whereas their ablation effect is smaller; this distinction indicates sensitivity without equivalent causal necessity. Pose matters mainly for “perf,” and the final “ume” subtoken is comparatively insensitive to all four coarse removals. The agreement between hand saliency and hand ablation supports a manually grounded decision in this example, but one selected sequence cannot establish a dataset-wide anatomy claim.

## 7.2 Can Gemma repair token errors?

Table 2 answers this question for the current isolated-label regime. At zero corruption, Gemma preserves exact content in 95% of rows and keeps chrF near 100. At every tested nonzero rate, exact accuracy is already zero before correction and remains zero afterward. chrF decreases rather than improves. Thus, no nonzero error tolerance is demonstrated: there is no tested nonzero corruption level at which the language stage recovers exact content. Because the short sequences quantize every nonzero nominal rate to at least one substitution, this experiment does not locate a continuous threshold between 0 and 0.10.

This is not evidence that language models cannot correct sign transcription in general. A substitution such as “photo”  $\rightarrow$  “green” in an isolated label is compatible with no surrounding syntax or semantics that identifies the original word. The appropriate next experiment is therefore not another prompt sweep on LSA64, but a pre-registered sentence corpus in which controlled token errors leave genuine contextual redundancy.

**Table 2.** Gemma 3n E2B correction audit on 120 isolated-label rows per condition. chrF is on a 0–100 scale. On these short sequences, every nonzero nominal rate produces at least one substitution.

| Substitution rate | Exact before | Exact after | chrF before | chrF after | $\Delta$ chrF |
|-------------------|--------------|-------------|-------------|------------|---------------|
| 0.00              | 1.000        | 0.950       | 100.00      | 99.16      | -0.84         |
| 0.10              | 0.000        | 0.000       | 36.67       | 34.58      | -2.10         |
| 0.20              | 0.000        | 0.000       | 36.67       | 34.58      | -2.10         |
| 0.30              | 0.000        | 0.000       | 36.67       | 34.58      | -2.10         |
| 0.40              | 0.000        | 0.000       | 34.29       | 32.35      | -1.94         |
| 0.50              | 0.000        | 0.000       | 23.30       | 21.85      | -1.45         |

## 8 Discussion

The main result is an interface result. Predicting tokenizer IDs turns the original embedding-imitation idea into an explicit communication channel between the visual encoder and the LLM. It also separates two sources of success: IMITATOR must first place the correct IDs on that channel; only then can Gemma contribute language knowledge. Fluency from the downstream model cannot conceal an incorrect visual sequence because upstream exact and edit metrics are reported independently.

The current evidence does not yet support the strongest product story—that Gemma repairs imperfect sign transcription. It supports the prerequisite: sign streams from held-out signers can be mapped directly into ordered, valid Gemma IDs with substantially higher sequence accuracy than the initial decoder. The negative corruption audit explains why correction is absent here and produces a concrete data requirement for the next stage: natural multiword sentences, multiple plausible contexts, and enough redundancy that a corrupted token is inferable.

Discrete IDs trade flexibility for auditability. Unlike continuous prompts, they do not communicate a calibrated posterior over alternative tokens. Future work should compare top- $k$  or posterior-weighted interfaces, as explored in speech prompting [3,11], while retaining a separately scored visual transcript. Such uncertainty may give the LLM useful alternatives without returning to unconstrained embedding imitation.

## 9 Limitations and Scope

The study uses synthetic concatenations of isolated LSA64 clips, not naturally continuous signing. It contains no coarticulation, natural sentence order, or open-vocabulary text; the 121-row output layer covers only IDs observed in this task. Labels are Spanish lexical annotations of LSA signs, so direct token prediction is not equivalent to modeling LSA grammar. Skeleton inputs omit visual appearance and remain dependent on pose-estimation quality; isolated-sign findings cannot simply be transferred to continuous signing [9,13].

Only seed 23 is used, between-seed variance is unknown, and only two outer signers are prospectively confirmatory. Sample-level bootstrap intervals do not replace signer-level replication. E1 and the H2 threshold were developed on folds 4–6. The tokenizer restriction is efficient for a closed label set but must be expanded and revalidated for sentence data. Finally, the correction audit tests one model and a fixed prompt; its defensible conclusion is limited to the tested setup and rates, although the absence of contextual information is a structural limitation of the present data.

## 10 Conclusion

IMITATOR keeps the original goal of learning a lightweight bridge from sign motion to a pretrained language model, but replaces approximate embedding imitation with exact token-ID prediction. This change makes the bridge directly decodable, measurable, and compatible with an unchanged Gemma text interface. Controlled signer-held-out experiments show a substantial improvement in ordered token transcription. The downstream audit also prevents an overclaim: Gemma does not repair any tested nonzero token corruption in isolated labels. The next decisive test is sentence-level sign transcription, where linguistic context may turn the currently zero error budget into a useful correction channel.

## Reproducibility and Ethics

The confirmation configuration, metrics, perturbation seeds, and decision rules were committed before observing E1 outer results for signers 7–8. Source and checkpoint hashes, sample predictions, paired comparisons, and bootstrap outputs are stored in machine-readable artifacts. LSA64 is distributed for research under a CC BY-NC-SA license and its source describes ten non-expert subjects wearing colored gloves [15]. Public documentation inspected for this draft does not provide sufficient demographic or consent detail for broader fairness claims. Although keypoints reduce direct appearance exposure, motion traces remain biometric data and should not be treated as anonymous by default.

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